

Jonathan Kim

Madison, WI | jkim17.com | jonthankim1717@gmail.com | (608) 334-2501 | linkedin.com/in/jonthankim717 | github.com/JKim171

EDUCATION

University of Wisconsin-Madison, School of Letters and Science

May 2027

- **Bachelor of Science**, Majors in Computer Science and Data Science, Minor in Photography | GPA: 3.85/4.00
- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Data Science Modeling, Linear Algebra, Discrete Math, Computer Engineering, Machine Organization and Programming, Advanced Statistics

EXPERIENCE

Software Engineer Intern — Capital One, McLean, VA

May 2026 – Present

- Design and develop a self-service web application that automates the provisioning and management of enterprise security groups.
- Develop APIs that unlock previously siloed network security data, reducing cross-team data retrieval time by 90%.
- Implement automated workflows to validate new firewall rules against corporate security policies.
- Engineer an agentic Claude Skill to automate microservice scaffolding, reducing initial setup time by 50%.

Software Engineer Intern — Direct Supply, Milwaukee, WI

May 2024 – Aug 2024, May 2025 – Aug 2025

- Engineered and deployed end-to-end workflows for a web platform that increased user operational efficiency by 200%.
- Built client-facing features using React and RTK Query, enabling seamless access to new functionality and real-time data.
- Developed mobile features in Kotlin and Swift for Android and iOS, driving a 15% boost in engagement across over 50,000 users.
- Optimized SQL stored procedures, improving efficiency by 50% and building resilience.

Full Stack Engineer — Szczykutowicz Lab, University of Wisconsin-Madison

Oct 2023 – Present

- Designed and implemented AWS architecture for lab environments, managing cloud-hosted databases and data storage solutions.
- Constructed databases using MySQL by implementing 5,000+ configurations for CT scanning protocols.
- Developed optimization workflows, reducing scan times and radiation doses by 50%.

PROJECTS

FocusFrame

Feb 2026 - Present

- Built a browser-based eye-tracking platform to measure and visualize viewer attention on video content using React and Vite.
- Implemented real-time gaze tracking with MediaPipe FaceLandmarker, mapping iris landmarks to screen coordinates.
- Developed heatmaps, attention timelines, and gaze replay with Recharts and Canvas2D, supporting multi-viewer comparison.

Carpool Optimizer

Feb 2025 - Mar 2025

- Developed an algorithm to optimize carpool routes, leading to a 30% decrease in total driving distance for assigned drivers.
- Improved time efficiency by 70% and reduced fuel consumption by 25% by minimizing unnecessary detours and mileage.
- Developed an automated workflow using Python, enabling UW Club Tennis to efficiently schedule rides for 100+ members daily.

Chess Assistant Analyzer Bot

Jun 2023 – Jul 2023

- Developed bot to analyze 100,000+ user-inputted chess games and openings.
- Designed GUI that supports features such as adjustable depth of analysis and detailed animations of opening moves.
- Formulated statistical algorithm, targeting weaknesses and strengths, resulting in 20% increase in personal play rating.

TECHNICAL SKILLS

Languages: Python, Java, C#, C, TypeScript, JavaScript, Kotlin, Swift, SQL, HTML, CSS, MATLAB, R

Frameworks: React, SwiftUI, .NET, Unity, JUnit

Cloud & DevOps: AWS (RDS, EC2), GCP (VM), Docker, Git, Linux, Windows

Tools & Technologies: Docker, JIRA, Redux, RTK Query, MySQL, Pandas

AI & Automation: Claude code, Claude Skills, GitHub Copilot, LLM Orchestration, Agentic Workflows, Prompt Engineering

AWARDS

Wisconsin-Dairyland Programming Competition — 1st Place out of 71 teams (2023)

UW-Madison Club Tennis — National Championship 11th Place (2025, 2026), Colorado Classic 3rd Place (2024).

Future Business Leaders of America Nationals — Computer Applications 5th Place (2023), Business Management 4th Place (2022).

State of Wisconsin Higher Educational Aids Board — Academic Excellence Scholarship Winner (2023).